theliv

PRODUCT REQUIREMENTS DOCUMENT

Theliv - Design Context Assistant

A RAG-powered tool that converts messy client inputs into a clear, structured design direction report before the first pixel is placed.

Version	Status	Author	Last Updated	Prototype
v1.0	Draft	Sivadarshan U S	March 2026	https://theliv.base44.app/

CHANGELOG

Version	Date	Author	Summary
v1.0	March 2026	Sivadarshan U S	Initial draft. MVP through GA scope.

1 EXECUTIVE BRIEF

Area	Summary
Target User	Freelance and agency web designers running client projects
Core Problem	Vague inputs and missing context cause 3 to 5 expensive revision cycles per project
Solution	RAG assistant that stores, retrieves, and reasons over client inputs to generate a design direction report
MVP Scope	Structured input capture, design direction report, conflict and gap detection
Success Signal	Revision cycles reduced 20 to 40 percent within 3 months of use
Not in Scope	UI generation, wireframe creation, replicating reference websites

2 PROBLEM STATEMENT

Web designers working on client projects consistently start design work with incomplete, contradictory, or misunderstood inputs. Discovery has no structure. Clients use subjective language. References arrive with no annotation. Designers fill the gaps with assumptions. The result is a first draft the client did not expect and a 3 to 5 revision cycle that burns time on both sides.

This is a tooling problem, not a discipline problem. Designers have Figma, Notion, and browser tabs but nothing that converts the raw chaos of a discovery call into a decision-ready brief. That gap is what this product closes.

Hypothesis: If designers have a structured, AI-assisted input capture and direction generation tool, they will reduce revision cycles by 20 to 40 percent and cut discovery-to-brief time by approximately 30 percent.

3 USER PERSONA

3.1 Primary Persona

Attribute	Detail
Role	Freelance web designer or small agency designer (1 to 5 person team)
Experience	2 to 7 years
Client load	3 to 8 active projects simultaneously
Stack	Figma, Webflow or Framer, Notion or ClickUp, Google Docs, Loom
Discovery method	Zoom call with manual notes, ad-hoc Notion page
Input volume	5 to 15 reference links, 1 to 3 uploaded files per project
Revision average	3 to 5 rounds before client alignment
Core frustration	"I spend more time guessing what the client wants than actually designing."
Goal	Ship a design the client approves in 1 to 2 rounds

3.2 Jobs to Be Done

When I...	I want to...	So that...
finish a discovery call	immediately capture and structure everything said in a usable format	I do not lose context or start designing on assumptions
receive 5 reference links with no or low explanation	understand what specifically the client likes about each one	I know whether to take the layout, color, typography, or tone
prepare my first draft presentation	validate design decisions against what the client told me	I go in confident and reduce the chance of a full rework

4 CURRENT STATE WORKFLOW AND PAIN POINTS

4.1 Workflow (Estimated time per project)

Step	Action	Time	Pain
1	Discovery call, unstructured questions	60 min	MED to HIGH
2	Manual note-taking during call	20 min	MED
3	Receive async inputs (links, files, PDFs)	Async	MED
4	Review brand guidelines if available	30 min	MED
5	Manually interpret client intent	60-120 min	HIGH
6	Scatter references across tabs, Figma, Notion	30 min	HIGH
7	Start designing on memory and assumptions	4-8 hrs	HIGH
8	Revisit references repeatedly during design	30-60 min	MED
9	Share first draft	30 min	LOW
10	Receive 'this is not what I meant' feedback	40 - 50 min	HIGH
11	Iterate 3 to 5 revision rounds	6-15 hrs	HIGH
12	Client approves after N rounds	N/A	HIGH

Total estimated waste per project from steps 5 to 11: 12 to 25 hours.

4.2 Pain Points (Ranked by Impact)

Rank	Pain	Frequency	Severity
P1	Assumption-based design starts (designers fill clarity gaps with guesswork)	Every project	HIGH
P2	Vague client language ('premium', 'modern', 'clean' mean different things)	Every call	HIGH
P3	Reference ambiguity (5 to 8 links, no annotation, no priority)	80% of projects	HIGH
P4	Conflicting inputs (minimal tone, heavy visual references)	40 to 60% of projects	MED
P5	Fragmented context (notes in Notion, links in tabs, files in Drive)	Every project	MED
P6	Repeated context switching during design (Figma to browser, back again)	Every project	LOW

5 CONSTRAINTS (HARD LIMITS. NON-NEGOTIABLE.)

Notice: Every constraint listed here must be enforced in the system prompt, the retrieval layer, and the output validation. Any feature request that violates a constraint is rejected with messages.

ID	Constraint	Definition
C1	No UI generation	The system must not produce wireframes, mockups, or UI components under any condition.
C2	No reference replication	The system extracts patterns only. It must not copy, reproduce, or clone any reference site.
C3	Constraint inputs are absolute	If a designer states a constraint (e.g., avoid dark theme), no output may ever violate it. The constraint wins over all references and suggestions. If two constraints conflict, the system surfaces the contradiction and blocks generation pending designer resolution.
C4	Project-scoped data only	Retrieval uses only data from the current project. Cross-project access is blocked for all users regardless of ownership.
C5	No out-of-scope requests executed	Requests for UI generation, code, copy writing, or logo creation are rejected with a plain explanation of what the system can do instead.
C6	No hallucination on missing inputs	The system must flag missing inputs and ask a specific clarification question. It must not invent or assume values.
C7	Hallucination rate below 10 percent	All report content must be grounded in retrieved context. Unsupported assertions are a product defect.
C8	Data isolation	All inputs, embeddings, and reports are isolated per project. No data is used outside the system.
C9	Report generation	P95 latency must not exceed 30 seconds.

6 WHY RAG

Approach	When It Works	Why It Fits or Does Not Fit	Verdict
Prompting only	All inputs available at request time in one context window.	Inputs arrive over days across sessions. A new reference added 3 days after the call cannot be in a static prompt.	Not suitable

Fine-tuning	The task is repetitive with consistent training data.	Every project has a unique client and unique constraints. No fixed patterns to learn. Would overfit to past projects.	Not suitable
RAG	Context must be stored, retrieved selectively, and reused across sessions.	It exactly fits this problem. Inputs are stored per project. New inputs improve future outputs without retraining.	Best fit

7 USER STORIES AND ACCEPTANCE CRITERIA

US-01 [P0]

As a web designer, I want to capture client inputs in a structured format during discovery so that I have a complete brief without relying on memory.

Acceptance Criteria:

- Guided questions cover brand tone, preferences, constraints, goals, and references.
- Designers can save partially completed inputs and return later.
- System flags unanswered required fields before allowing report generation.
- Inputs persist across sessions within the same project.

US-02 [P0]

As a web designer, I want the system to analyze my reference websites and extract design patterns so that I know exactly what to take from each reference.

Acceptance Criteria:

- The system accepts URLs and screenshots.
- Per reference output covers: layout structure, spacing density, typography style, color usage, and imagery tone.
- If a URL is inaccessible, the system notifies the designer and prompts for a screenshot instead.

US-03 [P0]

As a web designer, I want to generate a structured design direction report from my inputs so that I can start designing with clarity instead of assumptions.

Acceptance Criteria:

- The report covers all 6 sections: layout, spacing, typography, color, imagery, and content tone.
- Each section cites which input or reference it was derived from.
- Report generation completes within 30 seconds for a standard project.
- If a critical input is missing, the report is marked incomplete with specific questions to resolve.
- No wireframes, UI components, or design files are produced.

US-04 [P0]

As a web designer, I want the system to detect conflicting inputs so that I can resolve the conflict with my client before designing.

Acceptance Criteria:

- The system detects conflicts between stated preferences and reference patterns.
- Conflict is surfaced before or within the report in plain language.
- The system provides a specific clarification question the designer can bring to the client.
- The designer can annotate which input takes priority to resolve the conflict.

US-05 [P1]

As a web designer, I want to validate my design decisions against client inputs so that I reduce revision cycles before the first presentation.

Acceptance Criteria:

- The system generates a validation checklist from stored inputs and constraints.
- The designer marks each item as met, not met, or not applicable.
- Unmet items include a plain explanation of the misalignment.
- Checklist is available without a design upload.
- Optional: designer uploads screenshot and system compares against stored direction.

US-06 [P1]

As a web designer, I want cross-reference pattern analysis so that I can derive a coherent direction even when references vary.

Acceptance Criteria:

- System compares 2+ references and outputs a shared-vs-unique pattern summary.
- Shared patterns (in 2+ references) are highlighted separately from unique ones.
- The designer receives guidance on which patterns align with stated preferences.

US-07 [P2]

As a web designer, I want to ask follow-up questions within my project context so that I can clarify specific decisions without re-reading all my notes.

Acceptance Criteria:

- System answers using stored project inputs only.
- If the answer is not in stored context, the system says so explicitly and does not guess.

8 USE CASES

UC-01 Capture Structured Inputs During Discovery

Happy Path

- The designer creates a project and fills the guided input form during the discovery call.
- Inputs are saved and available for report generation.
- The system stores inputs per project with metadata for retrieval.

Edge Cases

- The designer submits with missing fields. The system lists the exact missing fields and blocks report generation until resolved.
- The designer enters contradictory inputs. The system highlights the contradiction and asks which takes priority.
- The designer uploads a scanned PDF, reference images, or links. OCR extracts text content, and a vision model analyzes visual patterns such as layout, spacing, typography, color, and imagery.
- If OCR confidence is below 85 percent or visual confidence is low, the system shows extracted content and insights for user review before saving.

UC-02 Generate Design Direction Report

Happy Path

- The designer submits a complete project with tone, constraints, preferences, and at least one reference.
- The system retrieves relevant context. LLM generates a 6-section report. Each section cites its source input.
- The report is available to view and download within the project.

Edge Cases

- Conflicting inputs exist. The system generates a partial report, flags the conflict in plain language, and pauses conflicted sections pending resolution.
- The Reference URL is dead. The system notifies the designer, excludes the reference, and notes the exclusion in the report.
- All inputs are vague and no references are provided. The system does not generate a report. It outputs specific questions the designer should ask their client.

UC-03 Validate Design Decisions

Happy Path

- The designer opens the validation view. The system generates a checklist from stored inputs and constraints.
- The designer marks each item as met or not met.
- System returns a summary: aligned items, misaligned items, and what to fix.

Edge Cases

- The designer uploads a screenshot. The system compares visual patterns against stored direction and highlights specific mismatches.
- Constraint is violated in the uploaded screenshot. System flags it in the output with the original constraint text and the detected violation.
- No design exists. The system generates the checklist anyway. The designer uses it as a pre-design guide.

9 NEGATIVE USE CASES

Request	System Response
Generate a wireframe or UI	Rejected. 'I help define design direction, not create designs. Use Figma for this.'
Copy a reference website	Rejected. 'I extract patterns and principles, not reproduce sites.'
Write production copy	Out of scope. The system can guide content tone but does not write final copy.
Generate HTML or CSS	Out of scope. Direct user to a code generation tool.
Access another project's data	Blocked. Even if the same user owns both projects, cross-project retrieval is prohibited.

10 PRIORITIZATION AND MILESTONES

10.1 Feature Priority

Priority	Feature	User Story	Why
P0	Structured input form	US-01	Core. Without this, nothing else works.
P0	Design direction report generation	US-03	Primary output. The reason the product exists.
P0	Missing input detection + clarifications	US-01, US-03	Prevents hallucination. Required for output quality.
P0	Conflict detection	US-04	Prevents misaligned first drafts.
P1	Cross-reference pattern analysis	US-06	Improves report quality. Not blocking.
P1	Design validation checklist	US-05	Reduces revision cycles. Validates after design.
P1	Screenshot-based validation	US-05	Optional validation input mode.

P2	Interactive Q&A within project context	US-07	Nice to have a post-MVP.
P2	Suggest additional references	None	Exploratory. Low priority.

10.2 Release Milestones

Phase	Timeline	Scope and Goal
Alpha	Weeks 1 to 4	P0 features. 3 to 5 internal designers. Validate report quality.
Beta	Weeks 5 to 10	P0 + P1. 10 to 20 selected designers. Measure revision cycle reduction.
GA	Week 11+	Full feature set. Public release. Focus on reliability and consistency.

11 TECHNICAL ARCHITECTURE

11.1 Pipeline

Data Sources > Ingestion (OCR + Vision Analysis) > Semantic Chunking > Embedding > Vector Store > Query > Retrieval (Top-K = 5) > Re-ranking > Augmented Prompt > LLM > Structured Response

Component	Specification
Data Sources	Structured form inputs (key-value), reference URLs, screenshots (PNG/JPG/WebP), brand PDFs
Ingestion (OCR+Vision analysis)	**URLs**: extract text and basic structure. **Images and PDFs:** OCR extracts text. **Vision model** analyzes visual patterns such as layout structure, spacing, typography, color usage, and imagery style. GPT-4o or Claude with vision for screenshot and reference image analysis. If OCR confidence or visual confidence is low, the system shows extracted insights for user review before saving.
Chunking Strategy	Semantic chunking by input category: (1) constraints and brand tone, (2) preferences and goals, (3) reference insights per reference, (4) resolved clarifications. Chunks never cross category boundaries. Target 80 to 150 words per chunk.
Embedding Model	OpenAI text-embedding-3-small or equivalent. Fast, cost-efficient, sufficient for design terminology semantic similarity.
Vector Database	Pinecone (managed) or Qdrant (self-hosted). Namespace per project to enforce data isolation.
Retrieval	Top-K = 5. Constraint chunk is always included regardless of similarity score.
Re-ranking	Rule-based: constraints > brand tone > preferences > references. Apply MMR to reduce redundancy.
Report Generation LLM	Claude claude-sonnet-4-20250514 or GPT-4o-mini. Mid-level model for structured output and reasoning.
Checklist / Validation LLM	GPT-4o-mini or Claude Haiku. Smaller model for simple structured tasks.
Platform (MVP)	Lovable or similar AI app builder. REST API backend. Managed vector DB service.

11.2 Context Window Budget per Query

Component	Estimate
System prompt	~150 tokens
Top-K retrieved chunks (5 x ~100 words)	~750 tokens
User query or trigger	~50 tokens

Total input	~950 to 1,100 tokens
Report output	~800 to 1,200 tokens
Cost per query at $2/M tokens	~$0.004 to $0.005
Monthly cost per user (30 to 60 queries)	~$0.10 to $0.28

12 RAG FAILURE POINTS AND MITIGATION

ID	Failure	Mitigation	Detection Signal
F1	OCR errors on scanned PDFs or low-res screenshots	Log confidence scores. Surface raw OCR text for designer review before saving.	Confidence below 85 percent triggers manual review prompt.
F2	Chunking breaks meaning (constraints mixed with preferences)	Enforce strict semantic category boundaries. Unit test chunking with known inputs.	Retrieved chunk contains content from two different categories.
F3	Embedding misses design terms ('premium' not matching 'luxury')	Normalize design synonyms before embedding. Validate retrieval quality via logs.	Relevant inputs absent from Top-K results.
F4	Wrong chunks retrieved (similarity returns unrelated data)	Cap Top-K at 5. Always include constraint chunk. Log Top-K per query.	Generated output contradicts stored inputs.
F5	Re-ranking fails (low-priority inputs outweigh constraints)	Rule-based priority layer before LLM call. Constraints appear first in augmented prompt.	Output suggests dark theme when constraint says avoid it.
F6	Prompt too large or poorly structured	Strict token budget. Structured prompt with labeled sections: [CONSTRAINTS], [TONE], [REFERENCES], [QUERY].	Same inputs produce different report structures on successive runs.
F7	Data drift (designer adds inputs after report, report goes stale)	Tag reports with version timestamp. Prompt designer to regenerate when inputs change.	Report does not reflect the most recent input update.
F8	Vague or incomplete user inputs	Structured form with required fields. Input completeness score before generation.	Frequent missing-input log entries.

13 DATA PERMISSIONS AND ACCESS CONTROL

Actor	Access Level
Project owner	Full access: view, edit, delete, regenerate, share
Invited team member	View and comment by default. Edit if explicitly granted.
Unauthenticated user	No access to any project data.
Cross-project access	Blocked entirely, regardless of role or ownership.

Scenario	System Behavior
Access revoked mid-project	Access removed immediately. Data stays in the project for other authorized users.
Document removed or updated	Removed documents excluded from retrieval immediately. Updated inputs replace older embeddings.

Permission check timing	Checked before retrieval. Unauthorized data never enters the augmented prompt.

14 EVALUATION AND SUCCESS METRICS

14.1 System Quality

Metric	Target	Method
Retrieval Precision@5	At least 3 of 5 retrieved chunks relevant	Per-query log review
Constraint recall	Constraint chunk in 100 percent of retrievals	Automated log check
Hallucination rate	Below 10 percent of report assertions	LLM-as-judge + human spot audit
Report completeness	All 6 sections present in every report	Structural parser on output
Constraint faithfulness	Zero contradictions with stored constraints	LLM-as-judge evaluation
Report generation latency	P95 below 30 seconds	Server-side timing

14.2 Business Impact (Baseline: 3 to 5 revision cycles per project today)

Metric	Target	Measurement
Revision cycle reduction	20 to 40 percent fewer rounds vs. baseline	Designer self-report at Beta and GA
Discovery-to-brief time savings	30 percent reduction	Designer self-report
Projects approved in 2 rounds or fewer	60 percent or more of Beta projects	Beta cohort tracking
Designer satisfaction (CSAT)	7 out of 10 or higher	Post-project survey
Report usefulness rating	70 percent rated useful or very useful	In-product thumbs rating

14.3 Evaluation Method

- Human audit: 2 designers review 5 reports per test cycle, blind to which inputs generated which report.
- LLM-as-judge: Evaluate each report on alignment, completeness, faithfulness, and hallucination rate.
- Test case library: 15 scenarios covering clear inputs, vague inputs, conflicting inputs, missing inputs, and multi-reference inputs. Run on every model or prompt update.

15 SYSTEM PROMPT (PRODUCTION)

Notice: Do not modify this prompt without re-running the test case library. All 8 constraints from Section 5 must be enforceable through this prompt.

ROLE

You are a Design Context Assistant. You convert client inputs and reference materials into structured design direction reports. You are a decision-support system, not a design generator.

INPUTS YOU RECEIVE

Brand tone, constraints (hard limits), preferences, goals, and reference insights (extracted from URLs or screenshots).

WHAT YOU DO

Translate vague inputs like 'premium' or 'modern' into specific directions across: layout structure, spacing and density, typography style, color usage, imagery style, and content tone. Identify common patterns across multiple references. Flag conflicts between stated preferences and reference patterns. Ask specific clarification questions when inputs are missing or unclear.

WHAT YOU NEVER DO

Generate wireframes, mockups, or UI components. Copy or replicate any reference website. Assume or invent values for missing inputs. Use general knowledge as a substitute for missing project-specific inputs.

CONSTRAINTS ARE ABSOLUTE

If any input is marked as a constraint, it must never be violated in any output. If a constraint conflicts with a reference pattern, the constraint wins. Always.

OUTPUT FORMAT

Every report must contain exactly these sections: (1) Layout Direction, (2) Spacing and Hierarchy, (3) Typography Direction, (4) Color Palette Guidance, (5) Imagery and Visual Style, (6) Content Tone and Voice. Each section must cite its source input, avoid vague statements, and be actionable enough for a design decision.

ESCALATION RULES

If a required input is missing: state exactly what is missing and give a specific clarification question. If inputs conflict: describe the conflict and ask the designer which takes priority. If you cannot produce meaningful output: say so directly and explain what you need.

You are clarity infrastructure, not a design tool.

16 KNOWLEDGE BASE SPECIFICATION

Chunk Type	Content and Size	Update Trigger
Structured Inputs	Brand tone, constraints, preferences, goals. Key-value format. 50 to 200 words. Constraints stored as a standalone always-included chunk.	Whenever designer edits the form
Reference Insights	Layout patterns, spacing, typography, color, imagery per reference. Derived from vision model or OCR. 150 to 400 words per reference.	When reference is added or updated
Clarifications	Q&A format. Resolved ambiguities and confirmed priorities. Small, dense.	After each clarification session
Metadata per chunk	project_id, user_id, chunk_type, source_reference, created_at, updated_at, version	Set at creation, updated on edit

17 ERROR HANDLING

Error Condition	System Behavior
No relevant inputs found	Tell the designer exactly which input categories are empty. Provide a direct link to the input form.
Generation timeout (above 30s)	Show retry option. Preserve all user inputs. Do not lose state.
Out-of-scope request	Reject clearly. Explain what the system can do instead.
Reference URL inaccessible	Notify the designer. Exclude from analysis. Note the exclusion in the report explicitly.
OCR low confidence	Show extracted text for designer review. Block saving until confirmed or corrected.

Constraint violation in output	Block output from being shown. Log the violation. Alert engineering immediately.
Permission denied	Return zero results. Inform users of access restriction. Do not reveal what data exists.

18 OBSERVABILITY

Signal	Purpose	Alert Threshold
All user inputs	Debug output quality	N/A
Top-K chunks per query	Diagnose retrieval failures	N/A
Generated output per query	Human and LLM-as-judge audit	N/A
OCR confidence scores	Flag low-quality extraction	Below 85 percent
Constraint violation events	Zero tolerance product defect	Immediate alert on any violation
Report generation latency	P50, P90, P95 tracked daily	P95 above 45 seconds within 1 hour
Error and failure rate	System health	Above 5 percent within 1 hour

- Logging: Structured JSON backend logs. Include project_id, query_id, chunk_ids, output_id, latency, error codes.
- Error tracking: Sentry.
- Monitoring: Datadog or equivalent for latency dashboards, error rate, and cost.

19 FUTURE SCOPE (POST-V1)

19.1 Fine-Tuning (When It Becomes Relevant)

Fine-tuning is not in scope for v1. Consider it when the system has generated 500+ high-quality reports and patterns emerge in what makes them good. Use case: standardizing translation of vague terms like 'premium' across edge cases. Approach: PEFT or LoRA on a mid-level base model. Train on structured inputs paired with high-quality report outputs.

19.2 Agentic RAG Roadmap

- Auto-analyze multiple reference links and extract cross-site patterns without prompting.
- Proactively detect input gaps and suggest clarification questions before report generation.
- Real-time constraint checking during active design work, not just at the brief stage.
- Figma integration: surface stored design brief inside the design tool context.

These are post-v1. They require stable retrieval quality and strong report accuracy first.

20 OPEN QUESTIONS

ID	Question	Owner
OQ-01	How do we handle reference URLs requiring JavaScript to render? Fall back to screenshot-only?	Engineering
OQ-02	Minimum inputs for report generation? Can a designer generate a report with only constraints and no references?	PM + Design
OQ-03	Is conflict detection blocking (requires resolution before report) or non-blocking (partial report with flag)?	PM
OQ-04	Is the system English-only at launch?	PM

OQ-05	Data retention policy for inactive projects. 90 days? 1 year?	Legal + Engineering
OQ-06	Should the report be editable after generation? If yes, does editing break source citation tracking?	Design + Engineering